Product Introduction

ZDH3070 是一款单芯片全集成的射频前端芯片（RF Front-end IC）。它集成了功放（PA）、低噪放（LNA）和射频开关（SW）。它采用典型 3.3V 或 5V 供电，TX 端最大饱和输出功率 25dBm，典型 RX 端噪声系数 1.6dB。输入、输出阻抗已经内部匹配到 50Ω，外围应用电路简洁。

ZDH3070 采用标准绿色无铅 QFN3x3-16 封装，具有很好的可靠性、经济性和极高的性价比。适用于提高 UWB 收发系统的灵敏度和传输距离扩展。

Typical application scenarios

- 精准实时定位系统
- 距离感知无线网络
- IEEE802.15.4-2011 UWB 收发器

Absolute Maximum Ratings

Parameter	Value
Storage Temperature	-65°C~+150°C
Operating Temperature	-55°C~+125°C
Operating Voltage	6V
DC Control Input Voltage	5V
TX RF Input Power	+15dBm
ANT Input Power	+10dBm

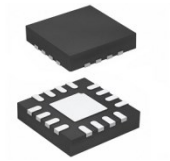
Operating status control logic table

State	TXEN	RXEN
TX on	H	L
RX on	L	H
Shutdown	L	L

"H": High level = +2.8V~VDD

"L": Low level = 0~+0.5V

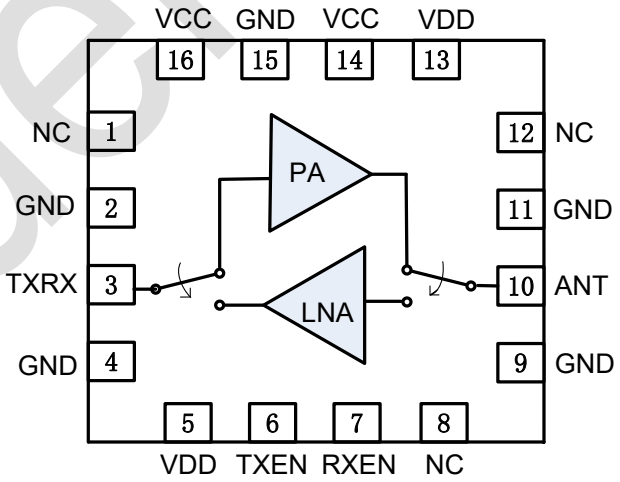
Features



- 3.3V 或 5V 单电压供电
- TX 小信号增益：21dB @ 5500MHz
- TX 饱和输出功率：22dBm @ 5500MHz
- TX/RX 端口复用，单一天线端口
- 单芯片 IC，集成了功放（PA）、低噪放（LNA）和两个单刀双掷开关
- 输入/输出 50Ω 阻抗匹配
- 绿色无铅 16 脚 QFN3x3 封装

本产品符合所有相关法规且不含卤素。

Pin diagram (Top View)



PIN No.	Pin name	Description
1,8,12	NC	Empty, floating or grounded
2,4,9,11,15	GND	Ground
3	TXRX	TXRX Transceiver RF signal nput/output
6	PEN	Transmit enable
7	LEN	Receive enable
10	ANT	Antenna RF signal input/output
5,13	VDD	RX Power supply voltage input
14,16	VCC	TX Power supply voltage input
17	EPAD	Bottom GND

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Recommended operating conditions

Parameter	Unit	Minimum	Typical	Maximum
Operating voltage	V	2.8	3.3	5
Control voltage "1"	V	2.8	3.0	VDD
Control voltage "0"	V	0	-	0.5

1. Test conditions: VDD=VCC=+3.3 V, Temp=+25° C, Freq: 4600MHz~6500MHz, PEN=3.0V, LEN=0V, 50Ω test system.

Parameter	条件	典型值					单位
频率范围(F)	-	4600	5000	5500	6000	6500	MHz
饱和输出功率(Psat)	Pin=5dBm	22	22	22	23	23	dBm
小信号增益(Gain)	Pin=0dBm	21	20	21	22	19	dB
输入回损 (S11)	-	-5	-4	-3	-6	-9	dB
输出回损 (S22)	-	-4	-3	-4	-12	-12	dB
P1dB	-	22	21	22	23	23	dBm
TX 静态电流 (Icq)	无 RF 输入信号	40	40	40	40	40	mA
TX 工作电流 (Icc)	-	199	224	237	206	177	mA
工作效率	-	16	14	14	19	22	%

2. Test conditions: VDD=VCC=+5V, Temp=+25° C, Freq: 4600MHz~6500MHz, PEN=3.0V, LEN=0V, 50Ω test system.

Parameter	条件	典型值					单位
频率范围(F)	-	4600	5000	5500	6000	6500	MHz
饱和输出功率(Psat)	Pin=5dBm	23	22	22	24	25	dBm
小信号增益(Gain)	Pin=0dBm	23	21	22	22	19	dB
输入回损 (S11)	-	-5	-3	-4	-7	-9	dB
输出回损 (S22)	-	-4	-3	-4	-15	-14	dB
P1dB	-	22	21	22	24	25	dBm
TX 静态电流 (Icq)	无 RF 输入信号	52	52	52	52	52	mA
TX 工作电流 (Icc)	-	217	227	241	241	223	mA
工作效率	-	17	12	13	22	29	%

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RX Electrical Parameters

1、测试条件：VDD=VCC=+3.3 V，Temp= +25°C，Freq：4600MHz~6500MHz，PEN=0V，LEN=3.0V，50Ω 测试系统。

Parameter	典型值					单位
频率范围(F)	4600	5000	5500	6000	6500	MHz
增益 (Gain)	26	24	21	19	18	dB
输入回损 (S11)	-3	-2	-2	-2	-3	dB
输出回损 (S22)	-8	-4	-3	-3	-4	dB
RX 静态电流 (I_{DQ})	21	21	21	21	21	dB
RX 工作电流 (I_{DD})	26	27	31	38	42	mA
P1dB	11	11	11	12	13	dBm
噪声系数	1.6	1.8	2.0	2.0	2.0	dB

2、测试条件：VDD=VCC=+5V，Temp= +25°C，Freq：4600MHz~6500MHz，PEN=0V，LEN=3.0V，50Ω 测试系统。

参数	典型值					单位
频率范围(F)	4600	5000	5500	6000	6500	MHz
增益 (Gain)	27	25	22	20	19	dB
输入回损 (S11)	-3	-2	-2	-2	-3	dB
输出回损 (S22)	-8	-4	-3	-3	-4	dB
RX 静态电流 (I_{DQ})	27	27	27	27	27	dB
RX 工作电流 (I_{DD})	46	39	48	52	49	mA
P1dB	15	15	15	15	14	dBm
噪声系数	1.6	1.8	2.0	2.0	2.0	dB

Chip shutdown electrical parameters

测试条件：VDD=VCC=+3.3 V，Temp= +25°C，PEN=0V，LEN=0V，50Ω 测试系统。

参数	条件	最小值	典型值	最大值	单位
关断后漏电流	-	-		10	uA
TXRX-ANT 隔离	-	-	-40	-	dB
ANT-TXRX 隔离	-	-	-40	-	dB
TX-RX 切换时间	10%RF-90%RF/ 90%RF-10%RF	-	300	-	ns
关断/开启切换时间	50%Vctrl-10%RF/50%Vctrl-90%RF	-	300	-	ns

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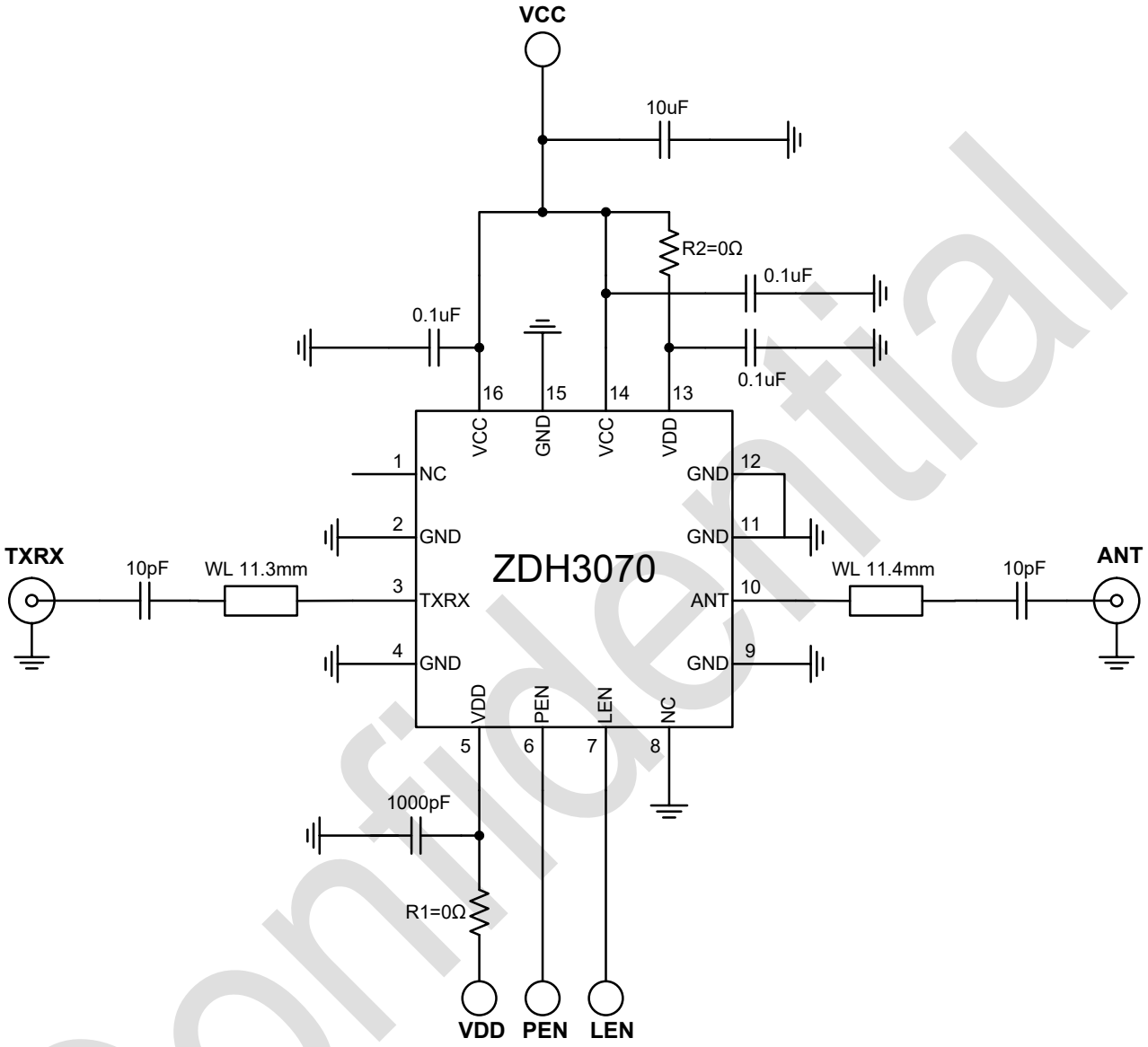
Application circuit diagram



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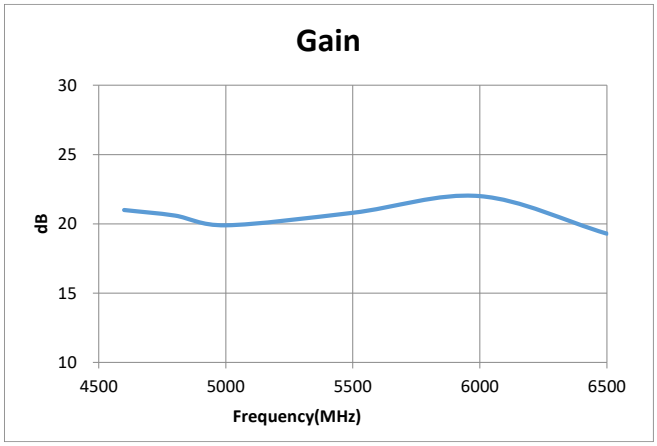
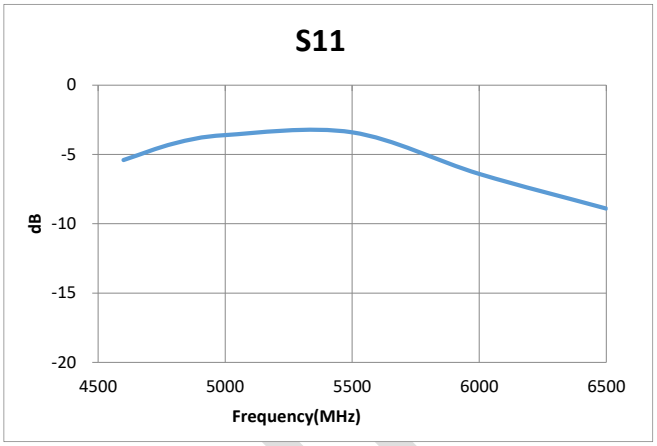
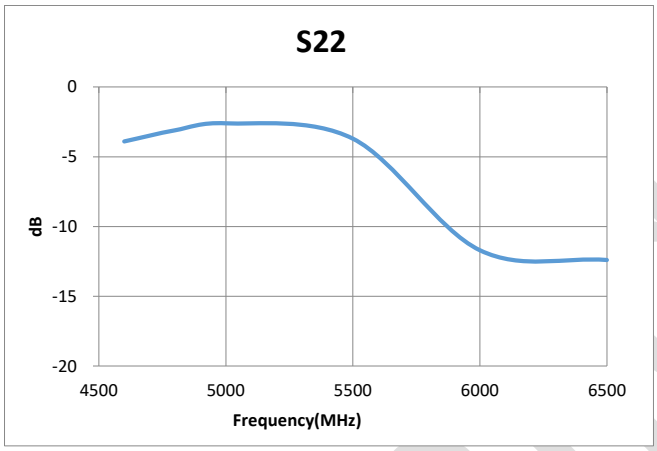
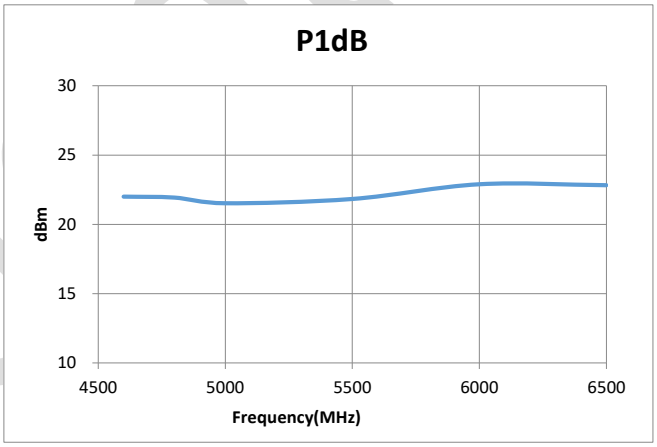
Typical performance curves (TX)

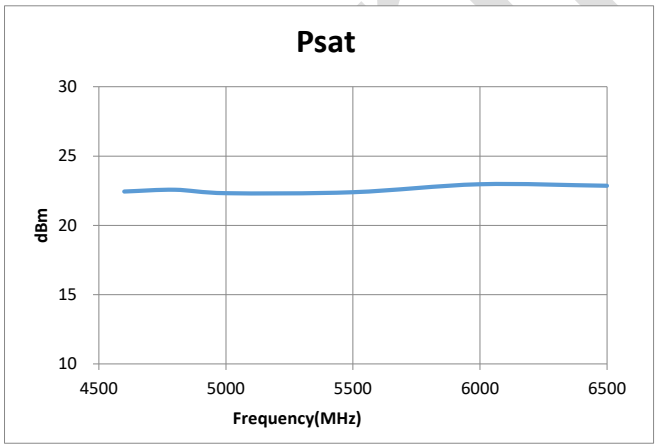
1. Test conditions: VDD=VCC=+3.3 V, Temp=+25° C, Freq: 4600MHz~6500MHz, PEN=3.0V, LEN=0V, 50Ω test system.

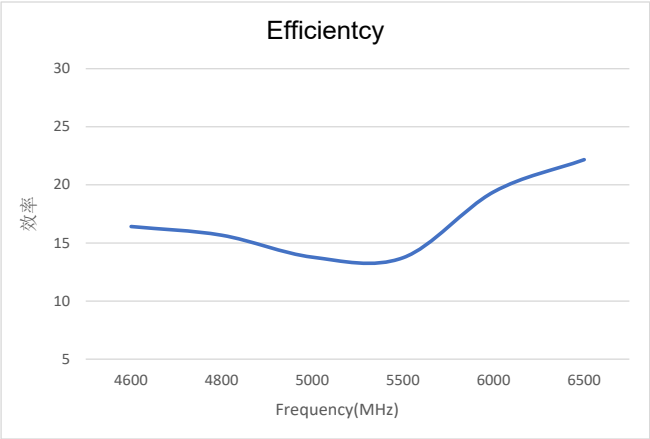






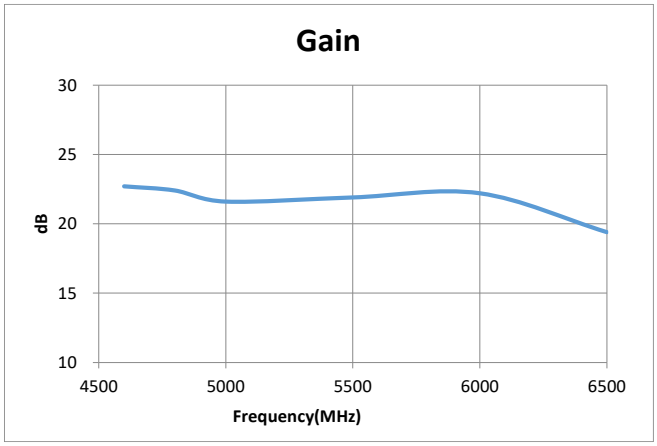


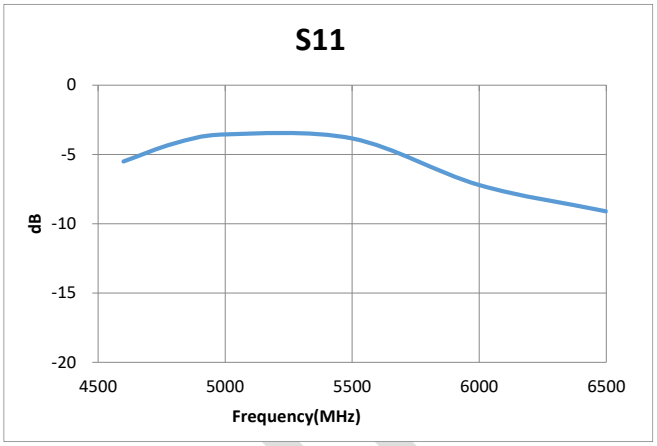


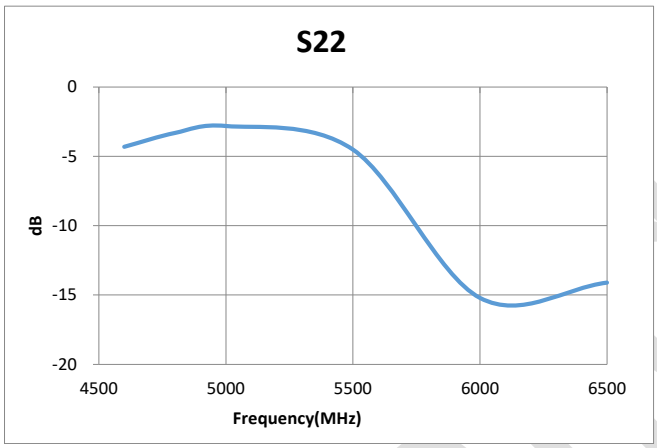


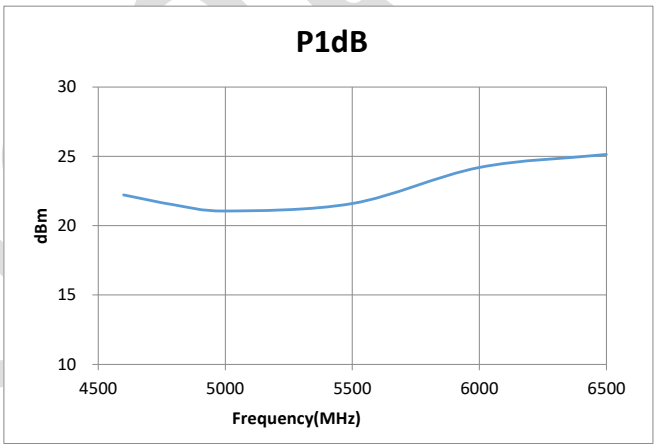
Typical performance curves (TX)

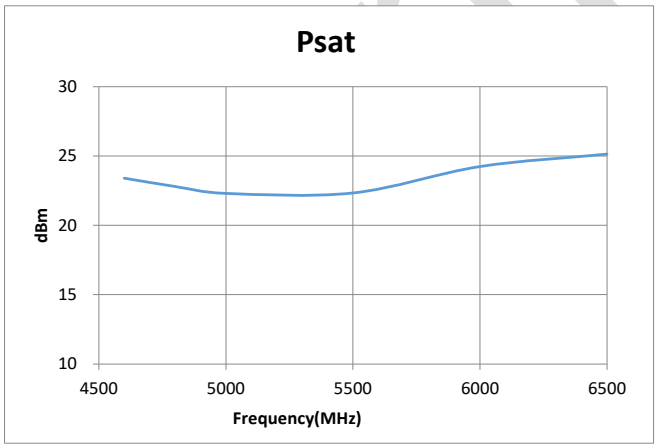
2. Test conditions: VDD=VCC=+5 V, Temp=+25° C, Freq: 4600MHz~6500MHz, PEN=3.0V, LEN=0V, 50Ω test system.

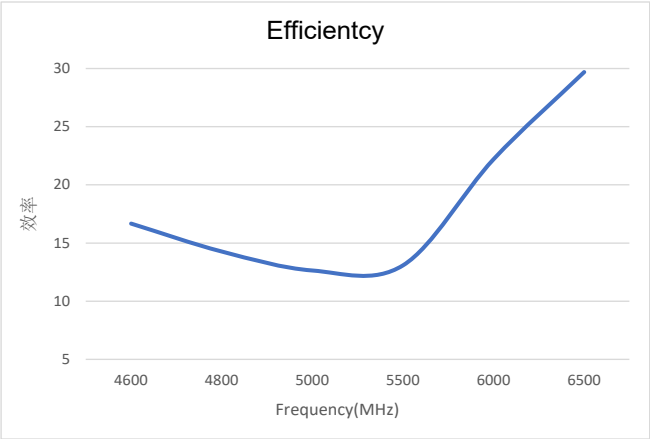






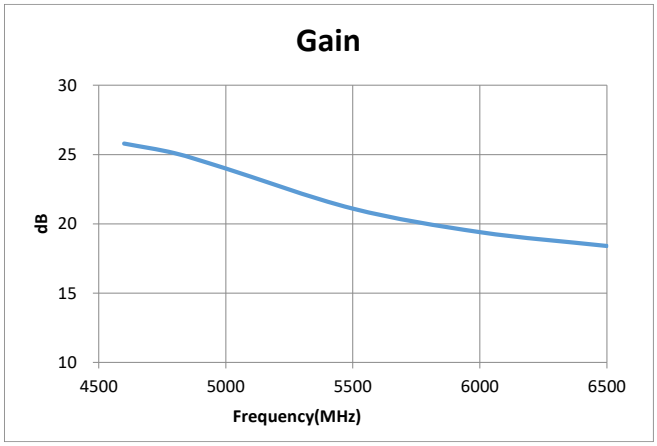


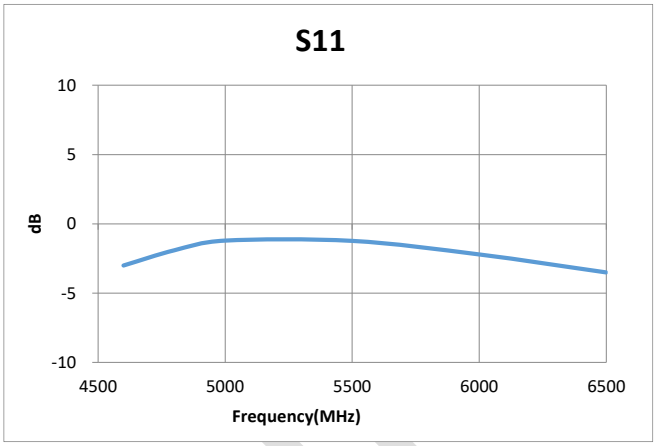


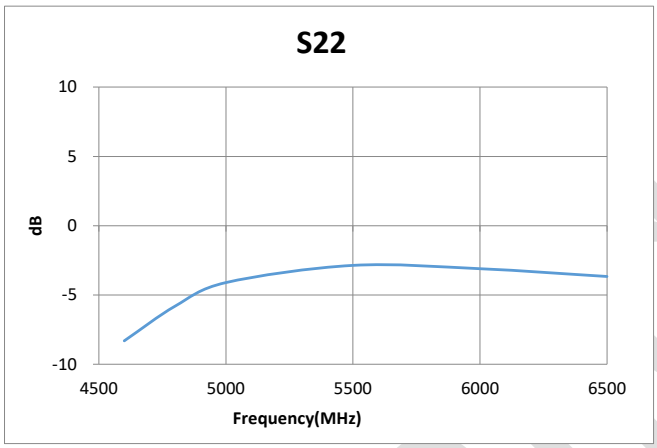


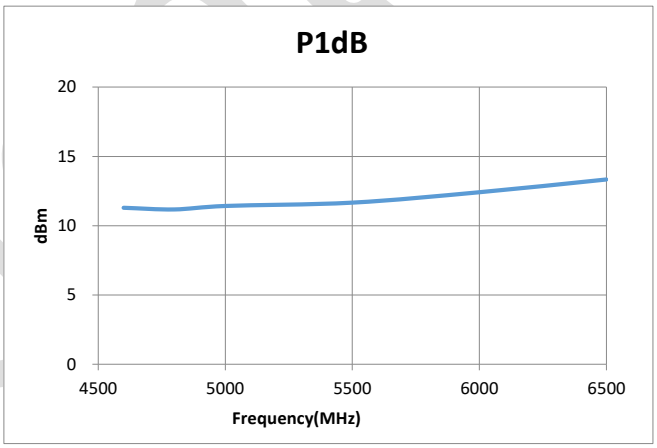
Typical performance curves (RX)

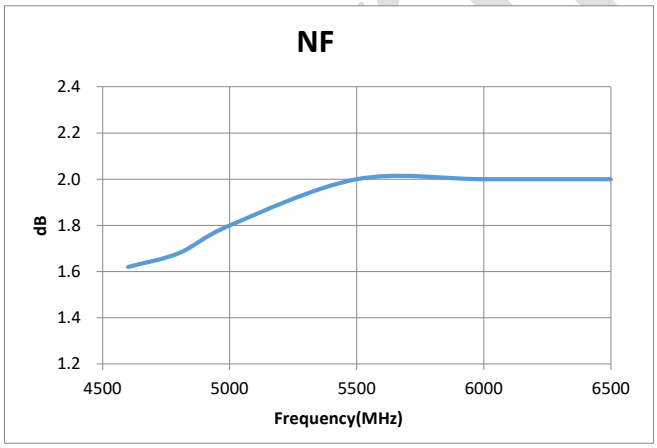
1. Test conditions: VDD=VCC=+3.3 V, Temp=+25°C, Freq: 4600MHz~6500MHz, PEN=0V, LEN=3.0V, 50Ω test system.





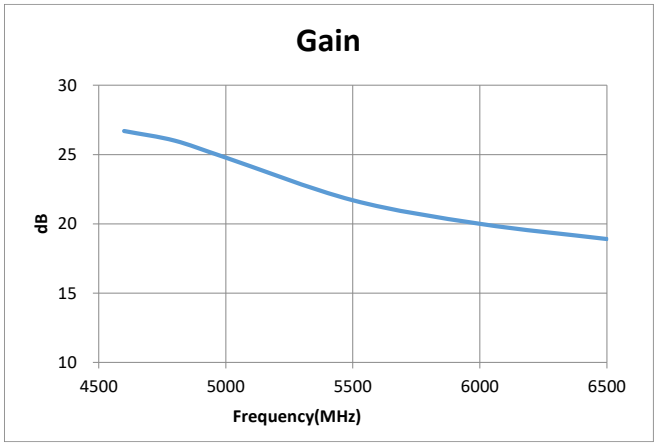


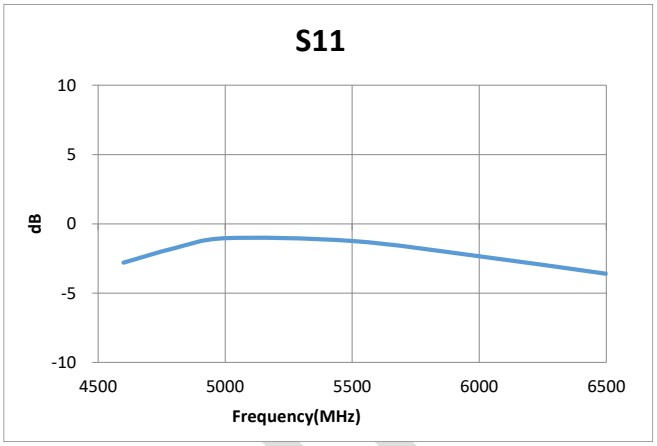


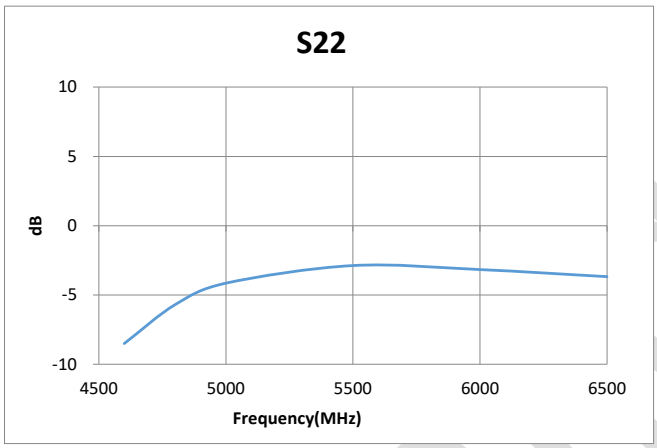


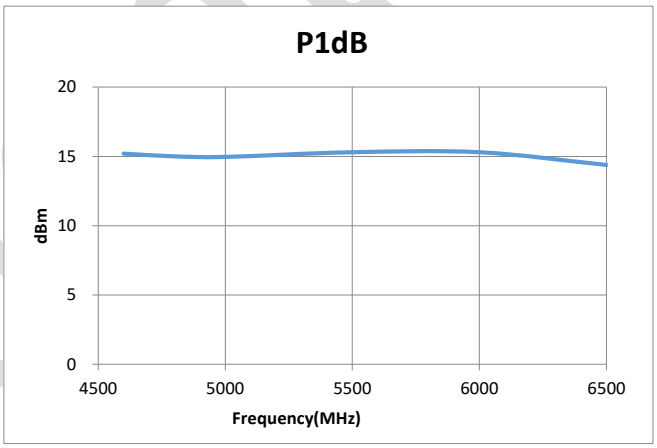
Typical performance curves (RX)

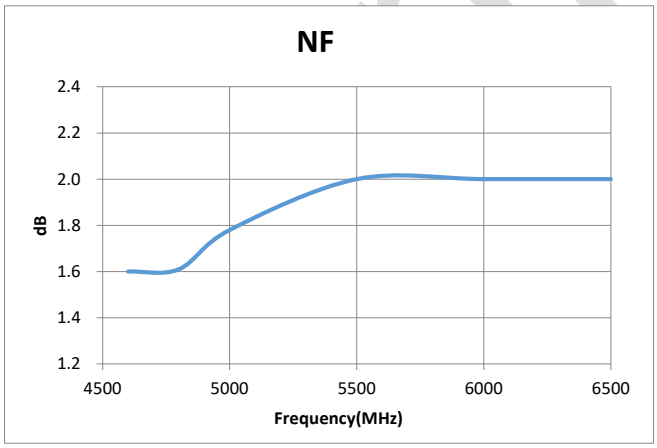
2. Test conditions: VDD=VCC=+5V, Temp=+25°C, Freq: 4600MHz~6500MHz, PEN=0V, LEN=3.0V, 50Ω test system.



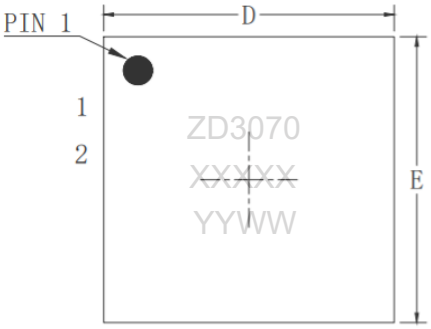




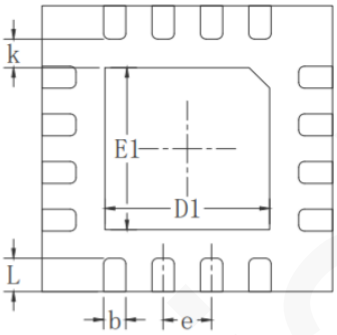




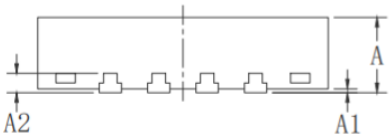
Package diagram



top view



bottom view



side view

symbol	Dimensions unit: mm		
	minimum value	Standard value	maximum value
A	0.700	0.750	0.800
A1	0.000	--	0.050
A2	0.195	0.203	0.211
D	2.950	3.000	3.050
E	2.950	3.000	3.050
D1	1.600	1.700	1.800
E1	1.600	1.700	1.800
k	0.300 Min.		
b	0.180	0.230	0.280
e	0.500 Typ.		
L	0.300	0.350	0.400

Ordering Information

Model	Marking	Packaging

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